

Full-Index Notation Considered Better

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Index-free notation is very popular. However, it fails to distinguish syntactically between types. This failure challenges static type checking of expressions like $\mathbf{v} + \mathbf{k}$. Is this a nonsensical addition of a vector to a 1-form, or, even worse, a vector added to a 2-form? In this note, we argue that, at least for symbolic computing and automated theorem-proving, the older, full-index notation is superior, being unambiguous yet not giving up much brevity.

In index-free notation, one typically writes contravariant vectors in bold Latin, as in $\mathbf{v}$, and covariant covectors in bold Greek, as in $\boldsymbol{\omega}$ (see, for instance, Part II of MTW¹). The argument for index-free notation is that geometric objects exist independently of particular choice of coordinates. However, I argue that such objects do not exist independently of *any* coordinates. One must ultimately choose *some* coordinates, and, no matter what coordinates one chooses, the dimensionality, or number of index values for each index, is the same. Once a coordinate system is chosen, one might as well denote a vector, for instance, by its coordinate tuple v^μ as by its index-free symbol $\mathbf{v}$. The tuple notation is not much longer, yet is totally unambiguous. There are further benefits to full-index notation (despite its being considered old-fashioned).² To summarize:

1. It honors the fact that *some* coordinates must be chosen.
2. It honors the fact that all coordinate systems have the same dimensionality; 4 in the case of relativistic spacetime.
3. It distinguishes contravariant (up) indices from covariant (down) indices, empowering type checking and tracking of *valence* (up-down pattern) of expressions.
4. Is is friendly to Einstein-summation, so a compiler³ can check expressions like $v_\mu = g_{\mu\nu}v^\nu$.

In the rest of this note, I present motivation about why tracking valence is critical, plus some handy mnemonics for some cases.

Valence: Contravariance and Covariance

Imagine two coordinate systems or *gauges* over spacetime: X , measuring distances in inches, and X' , measuring distances in feet. X is a set of real 4-tuples, $x^\mu = (x^0, x^1, x^2, x^3)$, and X' likewise, $x'^\mu = (x'^0, x'^1, x'^2, x'^3)$,⁴ with μ running from 0 through 3. Each is a copy of

¹ Charles W Misner, Kip S Thorne, and John Archibald Wheeler. *Gravitation*. Princeton University Press, 2017. ISBN 9780691177793

² David Lovelock and Hanno Rund. *Tensors, Differential Forms, and Variational Principles*. Dover Publications, New York, 1989. ISBN 978-0486658407

³ Brian Beckman. Relativity toolkit. <https://github.com/rebcabin/RelativityToolkit>, 2026

⁴ MTW put the prime on the index, as in x'^μ . Other authors, like us, put the prime on the coordinate symbol, as in x'^μ

$\mathbb{R}^4$.

In physics, tuples like x^μ or x'^μ can locate any point in spacetime, or can denote a measurement of the time and 3D spatial location of an event, with time measured in distance units.⁵

Let x^μ denote a measurement of the spacetime location of some event in X , in inches, and let x'^μ denote a measurement of the spacetime location of the same event in X' , in feet (mnemonic: remember that *the prime* is in *feet*). Dropping the index,⁶ one might say $x = 12 \times x'$ for any such matched pair of measurements.

Now imagine μ functions $x^\mu(x'^\nu)$ that, given a measurement tuple, x'^ν , in feet, produce components, x^μ , of a measurement of the location of the same event in inches.⁷

For the following arguments, we can drop the indices.⁸ Treating x as a coordinate-transformation function $x(x')$, we can compute its partial derivative,⁹ which, of course, is also a function of the same independent variable, x' . This partial derivative happens to be the constant, 12, in our example:

$$\frac{\partial x}{\partial x'} = 12 \quad (\text{inches per foot}) \quad (1)$$

We always insist that our coordinate-transformation functions be invertible and differentiable, so the inverse function $x'(x)$ exists and has the partial derivative:

$$\frac{\partial x'}{\partial x} = \frac{1}{12} \quad (\text{feet per inch}) \quad (2)$$

Now, imagine another quantity, v^μ , which measures velocity in the x , or ‘inches,’ gauge. We wish to know its value, v'^μ , in the ‘feet’ gauge. We know the answer, hardly without thinking!

$$\begin{aligned} v'^\mu (\text{feet per second}) &= v^\mu (\text{inches per second}) \left/ \frac{\partial x}{\partial x'} \right. \\ v'^\mu &= v^\mu \frac{\partial x'}{\partial x} \end{aligned} \quad (3)$$

This is the definitional example of a *contravariant quantity*. A way to remember it is

$$\left(\begin{array}{c} \text{quantity} \\ \text{measured in} \\ \text{primed} \\ \text{gauge} \end{array} \right) = \left(\begin{array}{c} \text{quantity} \\ \text{measured in} \\ \text{unprimed} \\ \text{gauge} \end{array} \right) \frac{\partial \left(\begin{array}{c} \text{primed} \\ \text{gauge} \end{array} \right)}{\partial \left(\begin{array}{c} \text{unprimed} \\ \text{gauge} \end{array} \right)} \quad (4)$$

i.e., the unprimed quantity on the right-hand side is multiplied by a derivative with the unprimed gauge in the *denominator*.

⁵ Measure time in distance units by multiplying time by the universal constant, c , the speed of light, which has the same values, one for each gauge, in all inertial frames:

$$\begin{aligned} c &\approx 1.18 \times 10^{10} \text{ in/s and} \\ c' &\approx 9.84 \times 10^8 \text{ ft/s.} \end{aligned}$$

⁶ ironically, or just imagining a 1-dimensional setting

⁷ We use the same symbol, x^μ , for the coordinates of the event and for the functions that convert measurements. This kind of ambiguity is common in mathematics.

⁸ ironically, again

⁹ In full flower, we would have the 16-element Jacobian matrix, $\partial x^\mu / \partial x'^\nu$

Now consider a quantity, u_μ , measuring temperature gradient in degrees per inch, and its partner in the other gauge, u'_μ , measuring the same temperature gradient, but in degrees per foot. Again, we know the transformation almost without thinking:

$$\begin{aligned} u'_\mu \text{ (degrees per foot)} &= u_\mu \text{ (degrees per inch)} \times \frac{\partial x}{\partial x'} \\ u'_\mu &= u_\mu \frac{\partial x}{\partial x'} \end{aligned} \quad (5)$$

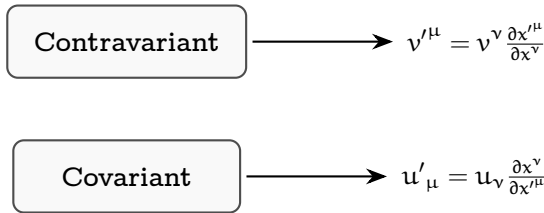
This is the definitional example of a *covariant quantity*. The way to remember it is

$$\left(\begin{array}{c} \text{quantity} \\ \text{measured in} \\ \text{primed} \\ \text{gauge} \end{array} \right) = \left(\begin{array}{c} \text{quantity} \\ \text{measured in} \\ \text{unprimed} \\ \text{gauge} \end{array} \right) \frac{\partial \left(\begin{array}{c} \text{unprimed} \\ \text{gauge} \end{array} \right)}{\partial \left(\begin{array}{c} \text{primed} \\ \text{gauge} \end{array} \right)} \quad (6)$$

i.e., the unprimed quantity on the right-hand side is multiplied by a derivative with the unprimed gauge in the *numerator*.

A contravariant quantity like velocity is not compatible with a covariant quantity like a gradient. Even though they both have four components, it makes no sense to add them. There are ways to transform one into the other, and ways to combine them via *contraction*, but those take us afar from the point of this short letter. The takeaway, here, is that we must distinguish them. If we're doing automatic computation, we should help the computer distinguish them. The best way to do that is symbolically, hence we conclude that explicit index notation is the way forward.

Finally, putting the indices back in, and assuming Einstein-summation (automatically sum over repeated indices in up-down pairs)¹⁰, we offer the following, final cheat-sheet:



¹⁰ The Relativity Toolkit's (op. cit.) compiler automatically performs such sums and automatically renames indices to prevent index collision and capture. Figure 1: Tensor transformation rules for contravariant and covariant indices.

References

Brian Beckman. Relativity toolkit. <https://github.com/rebcabin/RelativityToolkit>, 2026.

David Lovelock and Hanno Rund. *Tensors, Differential Forms, and Variational Principles*. Dover Publications, New York, 1989. ISBN 978-0486658407.

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